Pin Mapping - Tachysséma TD112 Board

FPGA Lattice LIFCL-40-8BG256I (CrossLink-NX) | Generated: 2026-10-25 | Living document (kept up to date)

Board reference

Development board	Tachysséma Développement TD112 (FPGA module + motherboard)
FPGA	Lattice LIFCL-40-8BG256I — CrossLink-NX family (Nexus platform), caBGA-256, speed grade 8 (High-Performance), Industrial temperature
Design tool	Lattice Radiant (Synplify Pro synthesis, VHDL-2008)
Clock inputs	100 MHz single-ended → D4 | 100 MHz LVDS → M16/M15 | 125 MHz LVDS → L7/M7 (all on PLL-dedicated / GPLL pins)
Reset button	SW2 → D12 (active-low, internal pull-up)
LEDs	RGB LED (WS2812B) → F15 | User LED → E9

Pmod connector pin map (A-G)

Each Pmod connector (A-G) has 12 pins with **zig-zag numbering**: odd pins (1,3,5,7,9,11) form one physical row, even pins (2,4,6,8,10,12) the other. Each row carries 4 GPIO + 1 GND + 1 VCC. In the `.pdc` use the **Ball/SITE** (`ldc_set_location -site {...}`); the **Pad** is Lattice's internal signal name. Power pins are not routed to the FPGA, so they have no Ball or Pad.

Conn.	Header	Pin	Ball / SITE (use in .pdc)	Pad (Lattice)	Function
A	J11	1	E10	PR4B	GPIO
A	J11	2	E12	PR6A	GPIO
A	J11	3	E13	PR10A	GPIO
A	J11	4	F9	PR3B	GPIO
A	J11	5	G15	PR20A	GPIO
A	J11	6	F10	PR8A	GPIO
A	J11	7	F16	PR19A	GPIO / LVDS pair (P)
A	J11	8	G16	PR19B	GPIO / LVDS pair (N)
A	J11	9	-	-	Power (GND)
A	J11	10	-	-	Power (GND)
A	J11	11	-	-	Power (+3V3)
A	J11	12	-	-	Power (+3V3)
B	J9	1	H14	PR27B	GPIO / LVDS pair (N)
B	J9	2	H11	PR30A	GPIO / LVDS pair (P)
B	J9	3	H15	PR27A	GPIO / LVDS pair (P)
B	J9	4	H10	PR30B	GPIO / LVDS pair (N)
B	J9	5	H13	PR32A	GPIO / LVDS pair (P)
B	J9	6	H16	PR26A	GPIO
B	J9	7	J11	PR36A	GPIO
B	J9	8	H12	PR32B	GPIO / LVDS pair (N)
B	J9	9	-	-	Power (GND)
B	J9	10	-	-	Power (GND)
B	J9	11	-	-	Power (+3V3)
B	J9	12	-	-	Power (+3V3)
C	J6	1	J12	PR36B	GPIO
C	J6	2	J15	PR34B	GPIO / LVDS pair (N)
C	J6	3	K11	PR38A	GPIO / LVDS pair (P)
C	J6	4	J16	PR34A	GPIO / LVDS pair (P)
C	J6	5	E6	PL4B	GPIO / LVDS pair (N)
C	J6	6	K12	PR38B	GPIO / LVDS pair (N)

Conn.	Header	Pin	Ball / SITE (use in .pdc)	Pad (Lattice)	Function
C	J6	7	D5	PL3B	GPIO
C	J6	8	D6	PL4A	GPIO / LVDS pair (P)
C	J6	9	-	-	Power (GND)
C	J6	10	-	-	Power (GND)
C	J6	11	-	-	Power (+3V3)
C	J6	12	-	-	Power (+3V3)
D	J5	1	K6	PL38A	GPIO
D	J5	2	E5	PL6B	GPIO / LVDS pair (N)
D	J5	3	E4	PL6A	GPIO / LVDS pair (P)
D	J5	4	F6	PL17A	GPIO / LVDS pair (P)
D	J5	5	G6	PL17B	GPIO / LVDS pair (N)
D	J5	6	H6	PL24A	GPIO / LVDS pair (P)
D	J5	7	H5	PL24B	GPIO / LVDS pair (N)
D	J5	8	J6	PL30A	GPIO
D	J5	9	-	-	Power (GND)
D	J5	10	-	-	Power (GND)
D	J5	11	-	-	Power (+3V3)
D	J5	12	-	-	Power (+3V3)
E	J12	1	F4	PL19A	GPIO / LVDS pair (P)
E	J12	2	J5	PL30B	GPIO
E	J12	3	G3	PL20A	GPIO
E	J12	4	L5	PL46B	GPIO / LVDS pair (N)
E	J12	5	G4	PL19B	GPIO / LVDS pair (N)
E	J12	6	L4	PL46A	GPIO / LVDS pair (P)
E	J12	7	J2	PL32B	GPIO
E	J12	8	K4	PL36A	GPIO
E	J12	9	-	-	Power (GND)
E	J12	10	-	-	Power (GND)
E	J12	11	-	-	Power (+3V3)
E	J12	12	-	-	Power (+3V3)
F	J7	1	K3	PL36B	GPIO
F	J7	2	J1	PL32A	GPIO
F	J7	3	H2	PL27B	GPIO / LVDS pair (N)
F	J7	4	H1	PL27A	GPIO / LVDS pair (P)
F	J7	5	N2	PL49A	GPIO / LVDS pair (P)
F	J7	6	H3	PL26B	GPIO / LVDS pair (N)
F	J7	7	N1	PL49B	GPIO / LVDS pair (N)
F	J7	8	H4	PL26A	GPIO / LVDS pair (P)
F	J7	9	-	-	Power (GND)
F	J7	10	-	-	Power (GND)
F	J7	11	-	-	Power (+3V3)
F	J7	12	-	-	Power (+3V3)
G	J8	1	M1	PL42A	GPIO / LVDS pair (P)
G	J8	2	M2	PL42B	GPIO / LVDS pair (N)
G	J8	3	L2	PL40B	GPIO / LVDS pair (N)
G	J8	4	K2	PL34B	GPIO / LVDS pair (N)
G	J8	5	L1	PL40A	GPIO / LVDS pair (P)
G	J8	6	K1	PL34A	GPIO / LVDS pair (P)
G	J8	7	M3	PL47A	GPIO / LVDS pair (P)
G	J8	8	L3	PL47B	GPIO / LVDS pair (N)

Conn.	Header	Pin	Ball / SITE (use in .pdc)	Pad (Lattice)	Function
G	J8	9	-	-	Power (GND)
G	J8	10	-	-	Power (GND)
G	J8	11	-	-	Power (+3V3)
G	J8	12	-	-	Power (+3V3)

Notes

• **Zig-zag numbering:** confirmed by the LVDS pairs in the pinout (a P/N pair can only sit on physically adjacent pins, which holds only under zig-zag numbering). Odd pins are one physical row, even pins the other.

• **Color per connector:** each connector has its own hue; rows alternate light/dark only for readability.

• **GPIO vs Power:** pins 1-8 are GPIO (routed to the FPGA). GND (9,10) and +3V3 (11,12) are power, identical on every connector, not routed to the FPGA.

• **Ball vs Pad:** the `.pdc` uses the Ball/SITE (e.g. K3); the Pad (e.g. PL36B) is Lattice's internal signal name.

• **LVDS:** pins marked 'LVDS pair' can be used as a differential pair (P/N) or as two independent single-ended GPIO.

• **Special functions** (PLL, I2C, PCLK, ADC): verify in Lattice Radiant's Device Constraint Editor (Dual Function column).

• **Data source:** pin map derived from the board pinout cross-checked against a reference project's constraints; treat as high-confidence, not exhaustively pin-by-pin verified.